Learning Monocular Gait Representations through Neurologically Guided Skeleton JEPA

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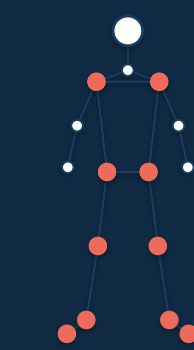


URTC 2026 · Research use only

SOURCE-VIDEO AUDIT

65.6% 6-fold CV · sequence IDs

24.0% 6-fold CV · source videos

−41.7 percentage points matched grouping gap

Source-video holdout changes the conclusion: 24.0% accuracy, below the 49.0% majority control.

1 · Research question

Can normal-only, label-free S-JEPA pretraining learn useful single-camera gait features, and do they survive a source-video holdout?

We trace the full path from RGB video to pose tokens, frozen embeddings, condition probes, source-identity audits, mask ablations, and causal latent forecasts.

2 · Small, source-concentrated cohort

GAVD subset

96 sequences **18 source videos**

Condition	Sequences	Videos
Normal	12	1
Parkinson's disease	9	2
Stroke	12	3
Cerebral palsy	16	2
Myopathic gait	47	10

All 12 normal clips come from one upload

This prevents a valid video-disjoint five-class estimate.

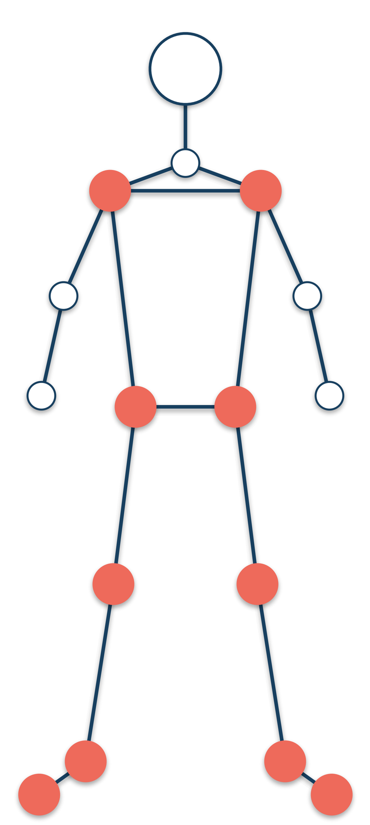
3 · Literature guides where to mask

Parkinson's
cadence · step length · support time

Stroke
laterality · temporal asymmetry

Cerebral palsy
flexed hip–knee–ankle chain

Myopathy
proximal weakness · trunk compensation



10 bilateral targets
shoulders · hips · knees ankles · foot indices

Interpretation boundary

These are video-level gait labels, not diagnoses. MediaPipe depth is not calibrated 3-D motion capture. Results do not measure transfer to a new person, camera, or clinic.

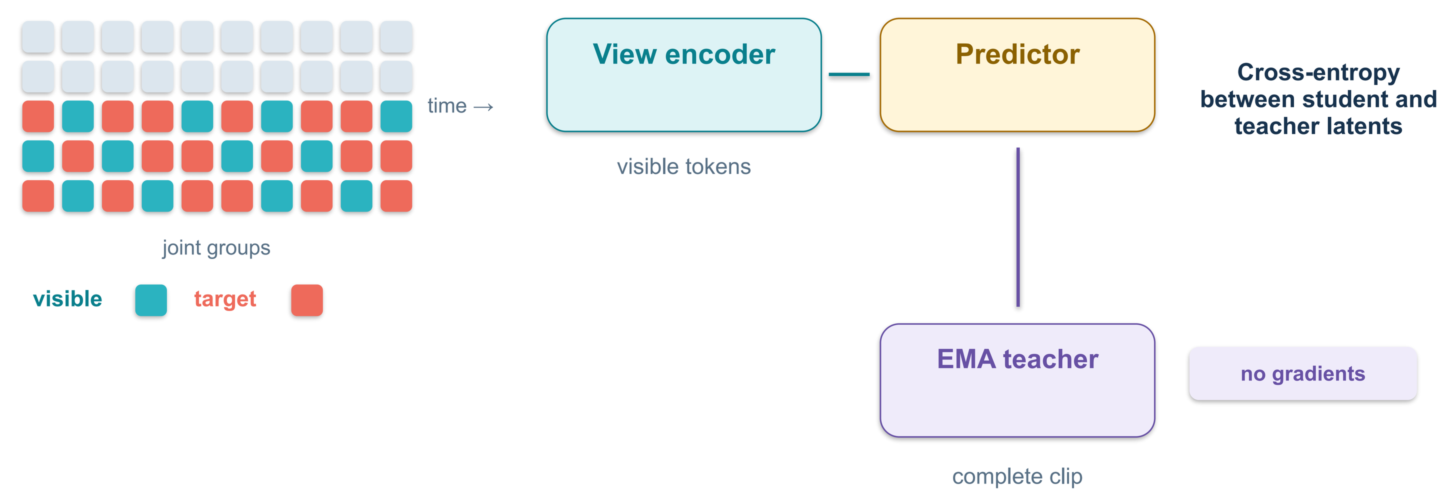
Feasibility study · Non-diagnostic · Transductive

4 · End-to-end audited pipeline

RGB clips	33 landmarks	64 frames	Mask ≤60%	384-d pool	5 classes
GAVD boxes	MediaPipe	528 tokens	eligible tokens	frozen encoder	Random Forest

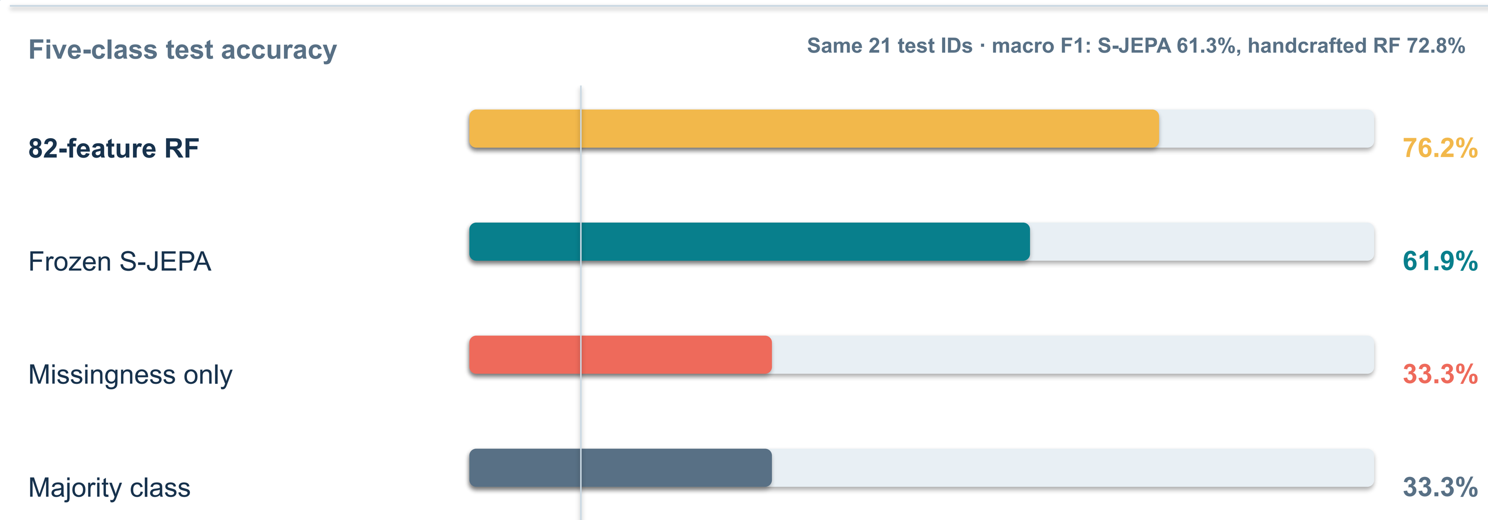
Pretraining uses 12 normal clips from one upload, no condition labels, 300 epochs, seed 42, and a frozen EMA target encoder.

5 · Predict a hidden representation, not coordinates



The view encoder sees context; the EMA teacher defines stable latent targets. Uniform target sampling avoids assuming that high motion is always clinically important.

6 · Exact 47/21 comparison



12.54 → 0.57 training loss

0.340 → 0.412 feature standard deviation

0.636 → 0.535 mean pair cosine

The learned representation is non-collapsed and useful for this split, but the comparison is between complete systems with different feature pipelines.

+28.6 points over missingness · −14.3 points vs handcrafted

7 · Source grouping changes the conclusion

Fixed 70/30 sequence split — video-confounded — **62.1%**

6-fold CV · sequence IDs — same fold machinery — **65.6 ± 11.7%**

6-fold CV · source videos — unseen uploads — **24.0 ± 16.5%**

41.7-point matched grouping gap
Video-grouped majority accuracy: 49.0%

8 · What did the representation retain?

Source identity — chance 5.6% · untrained 51.7% · raw pose 46.0% — **49.4 ± 12.1%**

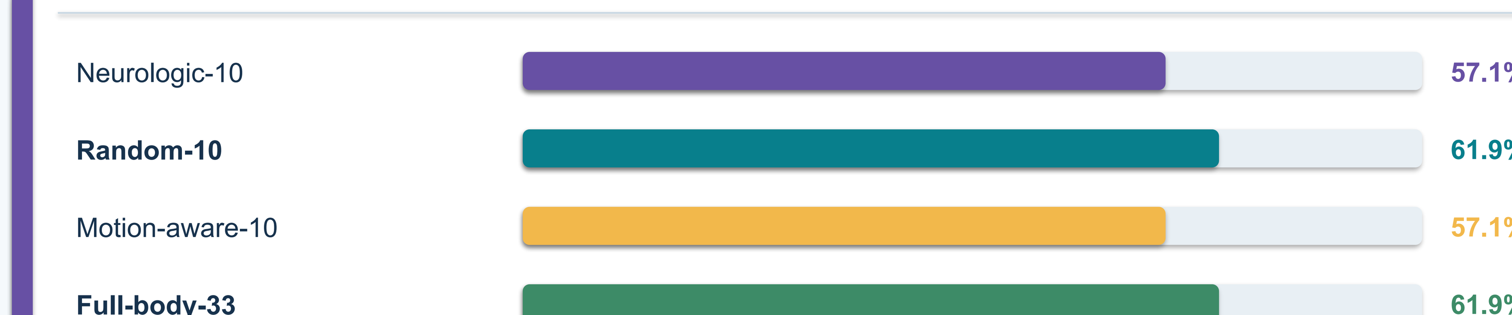
Best linear gait proxy — knee excursion; most canonical proxies near zero — **$R^2 = 0.086$**

Latent phase clock — grouped frame probe · null result — **$R^2 = -0.13$**

Causal forecast — normal 0.36 · abnormal 2.54–3.11 · source-confounded — **7.0–8.6×**

Source identity was already strong in untrained and raw-pose baselines. Most canonical gait proxies were not linearly readable.

9 · One-seed mask ablation: no clear winner



Same 21 test clips. One prediction separates 57.1% from 61.9%, so no arm earns a superiority claim.

Take-home message

At this scale, the evaluation unit changes the conclusion more than mask choice. The next valid test needs multiple normal source videos, grouped splits, uncertainty, and matched-compute baselines.

Audit the experimental unit first

Key sources: Zanardi et al., Sci Rep 2021 · Lauzière et al., IJPMR 2014 · Pandey et al., IJO 2023 · Maulet et al., Neurology 2023 · Assran et al., CVPR 2023 · Abdelfattah & Alahi, ECCV 2024 · Gil et al., GAVD 2020 · Kapoor & Narayanan, Patterns 2023 · Roberts et al., Ecography 2017

Code + reproducible artifacts
github.com/brilliantbeaver/alexpose/tree/main/penny/gavd3